DUCT LEAKAGE DIAGNOSTIC TEST

CALIFORNIA ENERGY COMMISSION

CEC-LMCV-MCH-20-H

SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS

CERTIFICATE OF VERIFICATION

Note: This table completed by ~~HERS~~ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. System Information

01	Space Conditioning System Identification or Name	
02	Space Conditioning System Location or Area Served	
03	Indoor Unit Name or Description of Area Served	
04	Building Type from LMCC	
05	Verified Low Leakage Ducts in Conditioned Space (VLLDCS) Credit from LMCC?	
06	Verified Low Leakage Air-Handling Unit Credit from LMCC?	
07	Duct System Compliance Category	
08	Any portions of Duct Located in Garage?	
09	Is the system type Small Duct High Velocity (SDHV)?	

B1. Duct Leakage Diagnostic Test for Completely New Duct System

01	Air-Handling Unit Airflow (AHU Airflow) Determination Method	
02	Condenser Nominal Cooling Capacity (ton)	
03	Indoor Unit Nominal Cooling Capacity	
04	Heating Capacity (kBtu/h)	
05	Conditioned Floor Area Served by this HVAC System (ft^2)	
06	Measured AHU Airflow (cfm)	
07	Duct Leakage Test Conditions	
08	Duct Leakage Test Method	
09	Leakage Factor	
10	Calculated Target Allowable Duct Leakage Rate (cfm)	
11	Actual Duct Leakage Rate from Leakage Test Measurement (cfm)	
12	Compliance Statement:	
13	Notes:	

B2. Duct Leakage Diagnostic Test for Low Leakage Ducts in Conditioned Space

01	System compliance with visual inspection per RA3.1.4.1.3?	
02	Duct Leakage Test Conditions	
03	Duct Leakage Test Method	
04	Target Allowable Duct Leakage Rate (cfm)	
05	Actual Duct Leakage Rate from Leakage Test Measurement (cfm)	
06	Compliance Statement:	

For information and data collection only. Not valid until registered with an ECC provider

B3. Duct Leakage Diagnostic Test for Low Leakage Air-Handling Unit (LLAHU)

01	Air-Handling Unit Airflow (AHU Airflow) Determination Method	
02	Condenser Nominal Cooling Capacity (ton)	
03	Indoor Unit Nominal Cooling Capacity	
04	Heating Capacity (kBtu/h)	
05	Conditioned Floor Area Served by this HVAC System (ft^2)	
06	Measured AHU Airflow (cfm)	
07	Duct Leakage Test Conditions	
08	Duct Leakage Test Method	
09	Leakage Factor	
10	Calculated Target Allowable Duct Leakage Rate (cfm)	
11	Actual Duct Leakage Rate from Leakage Test Measurement (cfm)	
12	Air-Handling Unit Manufacturer Name	
13	Air-Handling Unit Model Number	
14	Compliance Statement:	
15	Notes:	

For information and data collection only. Not valid until registered with an ECC provider

B4. Duct Leakage Diagnostic Test for Complete Replacement or Altered Duct System

01	Air-Handling Unit Airflow (AHU Airflow) Determination Method	
02	Condenser Nominal Cooling Capacity (ton)	
03	Indoor Unit Nominal Cooling Capacity	
04	Heating Capacity (kBtu/h)	
05	Conditioned Floor Area Served by this HVAC System (ft^2)	
06	Measured AHU Airflow (cfm)	
07	Duct Leakage Test Conditions	
08	Duct Leakage Test Method	
09	Leakage Factor	
10	Calculated Target Allowable Duct Leakage Rate (cfm)	
11	Actual Duct Leakage Rate from Leakage Test Measurement (cfm)	
12	Compliance Statement:	
13	Notes:	

B5. Duct Leakage Diagnostic Test for Smoke Test or Alteration Using Smoke Test

01	Air-Handling Unit Airflow (AHU Airflow) Determination Method	
02	Condenser Nominal Cooling Capacity (ton)	
03	Indoor Unit Nominal Cooling Capacity	
04	Heating Capacity (kBtu/h)	
05	Conditioned Floor Area Served by this HVAC System (ft^2)	
06	Measured AHU Airflow (cfm)	
07	Duct Leakage Test Conditions	
08	Duct Leakage Test Method	
09	Leakage Factor	
10	Calculated Target Allowable Duct Leakage Rate (cfm)	
11	Actual Duct Leakage Rate from Leakage Test Measurement (cfm)	
12	Compliance Statement:	
13	Notes:	

For information and data collection only. Not valid until registered with an ECC provider

For information and data collection only. Not valid until registered with an ECC provider

C. Ducts Located in Garage Spaces

01	Duct Leakage Test Method	
02	Leakage Factor	
03	Air-Handling Unit Airflow (AHU Airflow) Determination Method	
04	Measured AHU Airflow (cfm)	
05	Calculated Target Allowable Duct Leakage Rate (cfm)	
06	Actual Duct Leakage Rate from Leakage Test Measurement (cfm)	
07	Compliance Statement:	

D. Additional Requirements for Compliance

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	System was tested in its normal operation condition. No temporary taping allowed.	
02	Outside air (OA) duct connections to the central forced air duct system shall not be sealed/taped off during duct leakage testing. OA ducts used for Central Fan Integrated (CFI) Indoor Air Quality ventilation systems, or Central Fan Ventilation Cooling Systems, that utilize dampers that open only when OA is required and automatically close when OA is not required, may configure the OA damper to the closed position during duct leakage testing.	
03	All supply and return register boots were sealed to the drywall.	
04	Building cavities were not used as plenums, or platform returns, in lieu of ducts.	
05	If cloth backed tape was used it was covered with Mastic and draw bands.	
06	All connection points between the air handler and the supply and return plenums are completely sealed.	
07	For completely new systems visual inspection at final construction stage: For all supply and return registers, verify that the spaces between the register boot and the interior finishing wall are properly sealed.	
08	For completely new systems visual inspection at final construction stage: If the house rough-in duct leakage test was conducted without an air handler installed, inspect the connection points between the air handler and the supply and return plenums to verify that the connection points are properly sealed.	
09	For completely new systems visual inspection at final construction stage: Inspect all joints to ensure that no cloth backed rubber adhesive duct tape is used.	
10	For Duct Systems with Low Leakage Air-Handling Unit (LLAHU): The Low Leakage Air-handling Unit Model identified on this compliance document is included in the list of certified Low Leakage Air-Handling Units published on the Energy Commission Website at: https://www.energy.ca.gov/rules-and-regulations/building-energy-efficiency/manufacturer-certification-building-equipment/low	
11	For Replacement or Alteration Duct Systems: If the system complies using the Smoke Test method, the smoke test was conducted in accordance with the requirements of Reference Residential Appendix RA3.1.4.3.6. Systems that comply using the smoke test shall not be included in sample groups for ~~HERS~~ECC verification compliance.	
12	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
13	Correction Notes:	

E. Determination of ECC Verification Compliance

All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance.

01	

For information and data collection only. Not valid until registered with an ECC provider

For information and data collection only. Not valid until registered with an ECC provider

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Documentation Author Company Name:	Date Signed:
Address:	CEA/~~HERS~~ECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this certificate of installation is true and correct.
2. I am either: a) a responsible person eligible under division 3 of the business and professions code in the applicable classification to accept responsibility for the system design, construction, or installation of features, materials, components, or manufactured devices for the scope of work identified on this certificate of installation, and attest to the declarations in this statement, or b) I am an authorized representative of the responsible person and attest to the declarations in this statement on the responsible person’s behalf.
3. The constructed or installed features, materials, components or manufactured devices (the installation) identified on this certificate of installation conforms to all applicable codes and regulations and the installation conforms to the requirements given on the certificate of compliance, plans, and specifications approved by the enforcement agency.
4. I understand that a ~~HERS~~ECC rater will check the installation to verify compliance and if such checking determines the installation fails to comply, I am required to offer any necessary corrective action at no charge to the building owner.
5. I understand that a registered copy of this certificate of installation shall be posted or made available with the building permit(s) issued for the building and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
6. I understand that a registered copy of this certificate of installation is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to ensure this requirement is accomplished.

Responsible Builder/Installer Name:	Responsible Builder/Installer Signature:	
Company Name: (Installing Subcontractor or General Contractor or Builder/Owner)	Position With Company (Title):	
Address:	CSLB License:	
City/State/Zip:	Phone	Date Signed:
Third Party Quality Control Program (TPQCP) Status:	Name of TPQCP (if applicable):	

LMCV-MCH-20-H User Instructions

A. System Information

1 *HVAC System Identification or Name*: This field is filled out automatically. It is referenced from the LMCI-MCH-20, which must be completed prior to this document.
2. *HVAC System Location or Area Served*: This field is filled out automatically. It is referenced from the LMCI-MCH-20, which must be completed prior to this document.
3. *Indoor Unit Name:* This field is filled out automatically. It is referenced from the LMCI-MCH-20, which must be completed prior to this document.
4. *Building Type*: This field is filled out automatically. It is referenced from the LMCI-MCH-20
5. *Verified Low Leakage Ducts in Conditioned Space (VLLDCS)*: This field is filled out automatically. It is referenced from the LMCI-MCH-20.
6. *Verified Low Leakage Air-Handling Unit (VLLAHU) Credit* This field is filled out automatically. It is referenced from the LMCI-MCH-20.
7. *Duct System Compliance Category*: This field is filled out automatically. It is referenced from the LMCI-MCH-20.
8. *Any portions of Duct Located in Garage*: User select from Yes or No.

B1. Duct Leakage Diagnostic Test for Completely New Duct System

1. *Air-Handling Unit Airflow (AHU Airflow) Determination Method*: User will select from the following options:
   a. Default Airflow Method: The Default Airflow Method may only be used for homes where the duct system is being tested before the conditioning and heating system is installed and the equipment specification is not known (See Section RA3.1.4.2.1 of the 2025~~19~~ Reference Appendices).
   b. Cooling System Method: For systems with air conditioning, this selection must be made, and the nominal air handler airflow shall be 400 CFM per nominal ton of condensing unit cooling capacity as specified by the manufacturer
   c. Heating System Method: For heating only systems the nominal air-handler airflow shall be 21.7 CFM per kBtu/h of rated heating output capacity.
   d. Measured Airflow Method: The measured system airflow can be used as the air-handler airflow for the purpose of establishing duct leakage percentage (See Section RA3.1.4.2.3 of the ~~2019~~2025 Reference Appendices).
   e. Indoor Unit Method
2. *Condenser Nominal Cooling Capacity (ton)*: Same data given on MCH-01. Should be consistent with LMCI-MCH-20 for this system.
3. *Indoor Unit Nominal Cooling Capacity:* Same data given on MCH-01. Should be consistent with LMCI-MCH-20 for this system.
4. *Heating Capacity (kBtu/h)*: Same data given on MCH-01; Should be consistent with LMCI-MCH-20 for this system.
5. *Conditioned Floor Area Served by this HVAC System (ft^2)*: User must input CFA for the space. Should be consistent with LMCI-MCH-20 for this system.

For information and data collection only
Not valid until registered with an ECC provider

6. *Measured AHU Airflow (CFM)*: If "Measured Airflow Method" is selected, user must input measured airflow.
7. *Duct Leakage Test Conditions*: Select from the following options: Test Final is the only option for raters.
8. *Duct Leakage Test Method*: Select from the following options: Leakage to the Outside (house is pressurized simultaneously with the ducts such that only leakage going outside of the pressurized conditioned shell is measured, see RA3.1~~2~~.4.3.4), or Total Leakage.
9. *Leakage Factor*: This field is automatically filled out based on choices in previous fields.
10. *Calculated Target Allowable Duct Leakage Rate (cfm)*: This value will be automatically calculated based on values entered in previous fields.
11. *Actual Duct Leakage Rate from Leakage Test Measurement (cfm)*: Input the duct leakage rater taken from actual test measurements.
12. *Compliance Statement*: If Actual Duct Leakage Rate from leakage test is less than or equal to Calculated Target Allowable Duct Leakage Rate, "System passes leakage test" will automatically populate. If not, "System fails leakage test" will automatically populate.
13. *Notes*: This field is automatically filled out. The values in B02, B03, B04 and B05 are checked against the values in the same rows of the LMCI-MCH-20 for this system. If they do not match, an error message will appear here.

B2. Duct Leakage Diagnostic Test - Low Leakage Ducts in Conditioned Space

1. *System compliance with visual inspection per RA3.1.4.1.3~~2~~*: This field will be automatically filled. A visual inspection confirms the space conditioning system is located entirely in conditioned space in accordance with RA3.1.4.1.3. If any part of the duct system is outside of conditioned space, the system does not pass.
2. *Duct Leakage Test Conditions*: This field will be automatically filled. The entire duct system shall be included in the total leakage test. The air handler, supply and return plenums and all the connectors, transition pieces, duct boots and registers must be installed and tested to total system leakage. All supply registers shall be taped so that the tape goes over the grills and attaches to the surrounding drywall. All return grilles except for one large centrally located return grille or the air handler cabinet access panel shall be taped up.
3. *Duct Leakage Test Method*: This field will be automatically filled. Leakage to outside shall be verified by pressurizing the dwelling and the ducts to 25 Pa (0.1 inches of water) with respect to outside. A full description of these procedures can be found in RA3.1.4.3.4.
4. *Target Allowable Duct Leakage Rate (cfm)*: This field will be automatically filled. In order to pass this test duct leakage must be equal to or less than 25 cfm when the dwelling and ducts are pressurized to 25 Pa with respect to outside. NOTE: The 25 cfm leakage value will be difficult to reach unless the ducts are located in conditioned space.
5. *Actual Duct Leakage Rate from Leakage Test Measurement (cfm)*: Input the duct leakage rate taken from actual test measurements.
6. *Compliance statement:* This field will be automatically filled. The test passes if actual leakage rate is less than or equal to 25 cfm and a MCH-21 has been registered.

For information and data collection only. Not valid until registered with an ECC provider

B3. Duct Leakage Diagnostic Test - Low Leakage Air-Handling Unit (LLAHU)

1. *Air-Handling Unit Airflow (AHU Airflow) Determination Method*: User will select from the following options:
   a. Cooling System Method: For systems with cooling, this selection must be made, and the nominal air handler airflow shall be 400 CFM per nominal ton of condensing unit cooling capacity as specified by the manufacturer or the heating only value, whichever is greater (See Section RA3.1.4.2.2 of the 2025~~19~~ Reference Appendices).
   b. Heating System Method: For heating only systems the nominal air handler airflow shall be 21.7 CFM per kBtu/h of rated heating output capacity.
   c. Measured Airflow Method: The system airflow can be used as the air-handler airflow for the purpose of establishing duct leakage percentage (See Section RA3.1.4.2.3 of the 2025~~19~~ Reference Appendices).
   d. Default Airflow Method: The Default Airflow Method may only be used for homes where the duct system is being tested before the conditioning and heating system is installed and the equipment specification is not known (See Section RA3.1.4.2.1 of the 2025~~19~~ Reference Appendices).
   e. Indoor Unit Method
2. *Condenser Nominal Cooling Capacity (ton)*: Same data given on MCH-01.
3. *Indoor Unit Nominal Cooling Capacity:* Same data given on MCH-01.
4. *Heating Capacity (kBtu/h)*: Same data given on MCH-01;
5. *Conditioned Floor Area Served by this HVAC System (ft^2)*: User will input CFA for zone which should be consistent with the value from the LMCC. User will have the option to leave this field blank because the zone CFA is only required for the default airflow calculation.
6. *Measured AHU Airflow (cfm)*: If "Measured Airflow Method" is selected, user must input measured airflow.
7. *Duct Leakage Test Conditions*: User must select from the following options:
   a. Test Final: Test conducted at final inspection (testing at rough is not an option with this test. See Section RA3.1.4.3.1 of the 2025~~19~~ Reference Appendices).
8. *Duct Leakage Test Method*: User will select from the following options: Total Leakage.
9. *Leakage Factor*: Value will be automatically populated from in LMCI-MCH-20.
10. *Calculated Target Allowable Duct Leakage Rate (cfm)*: This value will be automatically populated depending on values in B06, B07, and B08.
11. *Actual Duct Leakage Rate from Leakage Test Measurement (cfm)*: User will input this value from actual measurements from leakage test.
12. *Air-Handling Unit Manufacturer Name*: This will be automatically populated from information entered in the MCH-01.
13. *Air-Handling Unit Model Number*: This will be automatically populated from information entered in the MCH-01.
14. *Compliance Statement*: If Actual Duct Leakage Rate from leakage test is less than or equal to Calculated Target Allowable Duct Leakage Rate, "System passes leakage test" will automatically populate. If not, "System fails leakage test will automatically populate.
15. *Notes:* This field is automatically filled out. The values in B02, B03, B04, B05, B12 and B13 are checked against the values in the same rows of the LMCI-MCH-20 for this system. If they do not match an error message will appear here.

For information and data collection only. Not valid until registered with an ECC provider

B4. Duct Leakage Diagnostic Test - Complete Replacement or Altered Duct System

1. *Air-Handling Unit Airflow (AHU Airflow) Determination Method*: User will select from the following options:
   a. Default Airflow Method: The Default Airflow Method may only be used for homes where the duct system is being tested before the conditioning and heating system is installed and the equipment specification is not known (See Section RA3.1.4.2.1 of the 2025~~19~~ Reference Appendices).
   b. Cooling System Method: For systems with air conditioning, this selection must be made, and the nominal air handler airflow shall be 400 CFM per nominal ton of condensing unit cooling capacity as specified by the manufacturer (Note: the heating only value may be used, if higher, See Section RA3.1.4.2.2 of the 2025~~19~~ Reference Appendices).
   c. Heating System Method: For heating only systems the nominal air handler airflow shall be 21.7 CFM per kBtu/h of rated heating output capacity.
   d. Measured Airflow Method: The measured system airflow can be used as the air handler airflow for the purpose of establishing duct leakage percentage (See Section RA3.1.4.2.3 of the 2025~~19~~ Reference Appendices).
   e. Indoor Unit Method
2. *Condenser Nominal Cooling Capacity (ton)*: Same data given on MCH-01. Should be consistent with LMCI-MCH-20 for this system.
3. *Indoor Unit Nominal Cooling Capacity:* Same data given on MCH-01.
4. *Heating Capacity (kBtu/h)*: Same data given on MCH-01. Should be consistent with LMCI-MCH-20 for this system.
5. *Conditioned Floor Area Served by this HVAC System (ft^2)*: User must input CFA for the space. Should be consistent with the LMCC input value. Should be consistent with LMCI-MCH-20 for this system.
6. *Measured AHU Airflow (CFM)*: If "Measured Airflow Method" is selected, user must input measured airflow.
7. *Duct Leakage Test Conditions*: Select from the following options:
   a. Test Rough-in AHU: Installers may determine duct leakage in new construction by using diagnostic measurements at rough-in building construction stage prior to installation of interior finishing (See Section RA3.1.4.3.2 of the 2025~~19~~ Reference Appendices). In this case the air handling unit (AHU) is installed at the time of test.
   b. Test Rough-in No AHU: Same as "Test Rough-in" except air handling unit is not yet installed (See Section RA3.1.4.3.2 of the 2025~~19~~ Reference Appendices).
   c. Test Final: Test conducted at "final", i.e. all equipment, ducts, and registers are installed and the system is essentially in its final operating condition. (rough-in no longer an option. See Section RA3.1.4.3.1 of the 2025~~019~~ Reference Appendices).
8. *Duct Leakage Test Method*: Select from the following options: Leakage to the Outside (house is pressurized simultaneously with the ducts such that only leakage going outside of the pressurized conditioned shell is measured, see RA3.1~~2~~.4.3.4), or Total Leakage.
9. *Leakage Factor*: This field is automatically filled out based on choices in previous fields.

For information and data collection only. Not valid until registered with an ECC provider

10. *Calculated Target Allowable Duct Leakage Rate (cfm)*: This value will be automatically calculated based on values entered in previous fields.
11. *Actual Duct Leakage Rate from Leakage Test Measurement (cfm)*: Input the duct leakage rater taken from actual test measurements.
12. *Compliance Statement*: If Actual Duct Leakage Rate from leakage test is less than or equal to Calculated Target Allowable Duct Leakage Rate, “System passes leakage test” will automatically populate. If not, “System fails leakage test” will automatically populate.
13. *Notes*: This field is automatically filled out. The values in B02, B03 and B04, B05 are checked against the values in the same rows of the LMCI-MCH-20 for this system. If they do not match an error message will appear here.

B5. Duct Leakage Diagnostic Test - Sealing All Accessible Leaks using Smoke Test

1. *Air-Handling Unit Airflow (AHU Airflow) Determination Method*: User will select from the following options:
    a. Default Airflow Method: The Default Airflow Method may only be used for homes where the duct system is being tested before the conditioning and heating system is installed and the equipment specification is not known (See Section RA3.1.4.2.1 of the ~~2019~~2025 Reference Appendices).
    b. Cooling System Method: For systems with air conditioning, this selection must be made, and the nominal air handler airflow shall be 400 CFM per nominal ton of condensing unit cooling capacity as specified by the manufacturer (Note: the heating only value may be used, if higher, See Section RA3.1.4.2.2 of the ~~2019~~2025 Reference Appendices).
    c. Heating System Method: For heating only systems the nominal air handler airflow shall be 21.7 CFM per kBtu/h of rated heating output capacity.
    d. Measured Airflow Method: The measured system airflow can be used as the air handler airflow for the purpose of establishing duct leakage percentage (See Section RA3.1.4.2.3 of the ~~2019~~2025 Reference Appendices).
    e. Indoor Unit Method
2. *Condenser Nominal Cooling Capacity (ton)*: Same data given on MCH-01.
3. *Indoor Unit Nominal Cooling Capacity:* Same data given on MCH-01.
4. *Heating Capacity (kBtu/h)*: Same data given on MCH-01.
5. *Conditioned Floor Area Served by this HVAC System (ft^2)*: User must input CFA for the space. Should be consistent with the LMCC input value.
6. *Measured AHU Airflow (CFM)*: If “Measured Airflow Method” is selected, user must input measured airflow.
7. *Duct Leakage Test Conditions*: Select from the following options:
    a. Test Rough-in AHU: Installers may determine duct leakage in new construction by using diagnostic measurements at rough-in building construction stage prior to installation of interior finishing (See Section RA3.1.4.3.2 of the ~~2019~~2025 Reference Appendices). In this case the air-handling unit (AHU) is installed at the time of test.
    b. Test Rough-in No AHU: Same as “Test Rough-in” except air handling unit is not yet installed (See Section RA3.1.4.3.2 of the ~~2019~~2025 Reference Appendices).

For information and data collection only. Not valid until registered with an ECC provider

    c. Test Final: Test conducted at "final", i.e. all equipment, ducts, and registers are installed and the system is essentially in its final operating condition. (rough-in no longer an option. See Section RA3.1.4.3.1 of the ~~2019~~2025 Reference Appendices).

8. *Duct Leakage Test Method*: Select from the following options: Leakage to the Outside (house is pressurized simultaneously with the ducts such that only leakage going outside of the pressurized conditioned shell is measured, see RA3.1~~2~~.4.3.4), or Total Leakage.
9. *Leakage Factor*: This field is automatically filled out based on choices in previous fields.
10. *Calculated Target Allowable Duct Leakage Rate (cfm)*: This value will be automatically calculated based on values entered in previous fields.
11. *Actual Duct Leakage Rate from Leakage Test Measurement (cfm)*: Input the duct leakage rater taken from actual test measurements.
12. *Compliance Statement*: If *Actual Duct Leakage Rate* is less than or equal to *Calculated Target Allowable Duct Leakage Rate*, "system passes - system complies with Allowable Duct Leakage Rate Criterion" will automatically populate.
    If measured leakage is greater than allowable duct leakage rate, then the following will automatically populate:
    "System passes using smoke test of an altered HVAC system in an existing building if:
    - No visible smoke exits the accessible portions of the duct system.
    - Smoke is only emanating from air handler unit AHU cabinet and non-accessible portions of the duct system.
13. *Notes*: This field is automatically filled out. The values in B02, B03, B04 and B05 are checked against the values in the same rows of the LMCI-MCH-20 for this system. If they do not match, an error message will appear here.

C. Ducts Located in Garage Spaces

1. *Duct Leakage Test Method*: This field is automatically filled out based on choices in previous fields.
2. *Leakage Factor*: This field is automatically filled out based on choices in previous fields.
3. *Air-Handling Unit Airflow (AHU Airflow) Determination Method*: This field is automatically filled out based on choices in previous fields.
4. *Measured AHU Airflow (CFM)*: This field is automatically filled out based on choices in previous fields.
5. *Calculated Target Allowable Duct Leakage Rate (cfm)*: This value will be automatically calculated based on values entered in previous fields
6. *Actual Duct Leakage Rate from Leakage Test Measurement (cfm)*: This field is automatically filled out based on choices in previous fields
7. *Compliance Statement*: If Actual Duct Leakage Rate from leakage test is less than or equal to Calculated Target Allowable Duct Leakage Rate, passes message will automatically populate. If not, "System fails leakage test" will automatically populate.

D. Additional Requirements for Compliance

1. This field must be a true statement (or not applicable) for the system to comply.
2. This field must be a true statement (or not applicable) for the system to comply.

For information and data collection only. Not valid until registered with an ECC provider

3. This field must be a true statement (or not applicable) for the system to comply.
4. This field must be a true statement (or not applicable) for the system to comply.
5. This field must be a true statement (or not applicable) for the system to comply.
6. This field must be a true statement (or not applicable) for the system to comply.
7. This field must be a true statement (or not applicable) for the system to comply.
8. This field must be a true statement (or not applicable) for the system to comply.
9. This field must be a true statement (or not applicable) for the system to comply.
10. This field must be a true statement (or not applicable) for the system to comply.
11. This field must be a true statement (or not applicable) for the system to comply.
12. *Verification Status:* If this Section does not apply, then select "All N/A". If the system meets all of the additional requirements for compliance that apply to the system, then select "Pass", otherwise select "Fail". The latter selection means that the system does not meet the applicable requirements and the system will need to be modified to meet the requirements or airflow and fan efficacy will have to be verified by diagnostic
13. *Correction Notes:* If one or more applicable requirements are not met "Fail" will appear in the row above. When this occurs the rater is required to enter detailed notes here that describe what failed and why.

Section E. Determination of ECC Verification Compliance

1. This field is filled out automatically. Compliance requires that all individual criteria pass.

Documentation Declaration Statements

1. The person who prepared the LMCV will sign and complete the fields for their name, company (if applicable), address, phone number, certification information (if applicable), date and signature.
2. The person who is assuming responsibility for the project being built to comply with Title 24, Part 6, will complete the fields (if applicable) for their company, responsible builder or installer name, CSLB license number, sample group number, dwelling test status in sample group, ECC Rater company name, ECC Rater name, ECC Rater signature, ECC Rater certification number and date signed.

For information and data collection only. Not valid until registered with an ECC provider

A. System Information

01	Space Conditioning System Identification or Name	<<text (data from LMCI-MCH-20)>>
02	Space Conditioning System Location or Area Served	<<text (data from LMCI-MCH-20)>>
03	Indoor Unit Name or Description of Area Served	<<text (data from LMCI-MCH-20)>>
04	Building Type from LMCC	<<text (data from LMCI-MCH-20)>>
05	Verified Low Leakage Ducts in Conditioned Space (VLLDCS) Credit from LMCC?	<<text (data from LMCI-MCH-20)>>
06	Verified Low Leakage Air-handling Unit Credit from LMCC?	<<text (data from LMCI-MCH-20)>>
07	Duct System Compliance Category	<<text (data from LMCI-MCH-20)>>
08	Any portions of Duct Located in Garage?	<<user entry, -Yes -No>>
09	Is the system type Small Duct High Velocity (SDHV)?	<<if the system type on the MCH-01= one of the following two: *small duct high velocity AC *small duct high velocity HP then value=yes; else value=no

B1. Duct Leakage Diagnostic Test New Duct System

<< If A07=New and A04 And A05 = false this section is required Else only display Section title and this note:
This section does not apply to this project.>>

01	Air-Handling Unit Airflow (AHU Airflow) Determination Method	<<pick one from list: DefaultAirflowMethod; CoolingSystemMethod; HeatingSystemMethod; MeasuredAirflowMethod; IndoorUnitMethod>>
02	Condenser Nominal Cooling Capacity (ton)	<< if B01 = CoolingSystemMethod, then user input is numeric x.xx; else =N/A >>
03	Indoor Unit Nominal Cooling Capacity (ton)	<< if B01 = IndoorUnitMethod, then user input is either numeric x.xx, else =N/A>>
04	Heating Capacity (kBtu/h)	<< if B01 = HeatingSystemMethod, then user input is numeric xxx.x; else =N/A >>
05	Conditioned Floor Area Served by this HVAC System (ft^2)	<<if B01 = DefaultAirflowMethod, user input is numeric xx,xxx; else = N/A (should be consistent with LMCI-MCH-20)>>
06	Measured AHU Airflow (cfm)	<< if B01 = MeasuredAirflowMethod, then user enter numeric x,xxx, else =N/A>>
07	Duct Leakage Test Conditions	<<only allowed value = TestFinal >>
08	Duct Leakage Test Method	<<user pick one from list: LeakageToOutside; TotalLeakage>>

09	Leakage Factor	<<calculated field:
		if TotalLeakage and MultiFamily and TestFinal then LeakageFactor=0.12;
		
		elseif LeakageToOutside and MultiFamily and TestFinal then LeakageFactor=0.06;
		
		else error message if invalid entries for arguments>>

10	Calculated Target Allowable Duct Leakage Rate (cfm)	<<calculated field: numeric xxx:
		if DefaultAirflowMethod then
		AHUAirflow=ZonedCondFloorArea*0.5* LeakageFactor;
		
		elseif AHUAirflowMethod= CoolingSystemMethod and A09 = no,
		then AHUAirflow=CondenserNomCoolCapacityTon*400* LeakageFactor;
		
		elseif AHUAirflowMethod = CoolingSystemMethod and A09=yes,
		then value=CondenserNomCoolCapacityTon *250*LeakageFactor;
		
		elseif AHUAirflowMethod= HeatingSystemMethod

For information and data collection only. Not valid until registered with an ECC provider

		then AHUAirflow=HeatingCapacityKbtuh*21.7* LeakageFactor;
		
		elseif AHUAirflowMethod= MeasuredAirflowMethod then AHUAirflow= Measured AHUAirflow * LeakageFactor;
		
		elseif AHUAirflowMethod= IndoorUnitMethod then AHUAirflow=IndoorAirUnitCoolingCapacityton*350*LeakageFactor >>

11	Actual Duct Leakage Rate from Leakage Test Measurement (cfm)	<<user input: numeric xxx.x>>
12	Compliance Statement:	<<if B11 measured leakage rate is ≤ to B10 target allowable leakage rate: "system passes leakage test"; else if measured leakage rate is > target allowable leakage rate: "system fails leakage test">>
13	Notes: <<if LMCI-MCH-20 row B02, B03, B04 and B05 ≠ LMCV-MCH-20 row B02, B03, B04 and B05, then display “The installed cooling capacity, heating capacity or CFA served does not match the Installation Certificate”, elseif LMCI-MCH-20 row B02, B03, B04 and B05 = LMCV-MCH-20 row B02, B03, B04 and B05, then display “ “>>	

B2. Duct Leakage Diagnostic Test for Low Leakage Ducts in Conditioned Space
<< If A07=New and 05=VLLDCS (true) this section is required Else only display Section title and this note:
This section does not apply to this project.>>

01	System compliance with visual inspection per RA3.1.4.1.3~~2~~?	<<text (data from LMCI-MCH-20)>>
02	Duct Leakage Test Conditions	<<text (data from LMCI-MCH-20)>>
03	Duct Leakage Test Method	<<text (data from LMCI-MCH-20)>>
04	Target Allowable Duct Leakage Rate (cfm)	<<text (data from LMCI-MCH-20)>>
05	Actual Duct Leakage Rate from Leakage Test Measurement (cfm)	<<user input: numeric xxx.x>>
06	Compliance Statement:	<<if B05 measured leakage rate is ≤ B04 target allowable leakage rate and B01 = complies with RA 3.1.4.1.3, then display text “System Passes Leakage Test”; else display text “System Fails Leakage Test”>>

B3. Duct Leakage Diagnostic Test for Low Leakage Air-Handling Unit (LLAHU)
<< If A07=New and 06=VLLAHU (true) this section is required Else only display Section title and this note:
This section does not apply to this project.>>

01	Air-Handling Unit Airflow (AHU Airflow) Determination Method	<<pick one from list: CoolingSystemMethod; HeatingSystemMethod; MeasuredAirflowMethod; DefaultAirflowMethod; IndoorUnitMethod>>
02	Condenser Nominal Cooling Capacity (ton)	<< if B01 = CoolingSystemMethod, then user input is numeric x.xx; else =N/A >>
03	Indoor Unit Nominal Cooling Capacity (ton)	<< if B01 = IndoorUnitMethod, then user input is either numeric x.xx, else =N/A>>
04	Heating Capacity (kBtu/h)	<<if B01 = HeatingSystemMethod, then user input is numeric xxx.x; else =N/A >>
05	Conditioned Floor Area Served by this HVAC System (ft^2)	<< if B01 = DefaultAirflowMethod, user input is numeric xx,xxx; else = N/A (should be consistent with LMCI-MCH-20)>>
06	Measured AHU Airflow (cfm)	<<if B01 = MeasuredAirflowMethod, then user enter numeric x,xxx, else = N/A>>
07	Duct Leakage Test Conditions:	<<text (data from LMCI-MCH-20)>>
08	Duct Leakage Test Method	<<text (data from LMCI-MCH-20)>>
09	Leakage Factor	<<text (data from LMCI-MCH-20)>>
10	Calculated Target Allowable Duct Leakage Rate (cfm)	<<calculated field: numeric xxx: if AHUAirflowMethod= CoolingSystemMethod and A09 = no,

For information and data collection only. Not valid until registered with an ECC provider

		then AHUAirflow=CondenserNomCoolCapacityTon*400* LeakageFactor;
		
		elseif AHUAirflowMethod = CoolingSystemMethod and A09=yes,
		then value=CondenserNomCoolCapacityTon *250*LeakageFactor;
		
		elseif AHUAirflowMethod= HeatingSystemMethod
		then AHUAirflow=HeatingCapacityKbtuh*21.7* LeakageFactor;
		
		elseif AHUAirflowMethod= MeasuredAirflowMethod then
		AHUAirflow= Measured AHUAirflow * LeakageFactor;
		
		elseif AHUAirflowMethod= DefaultAirflowMethod then
		AHUAirflow=ZonedCondFloorArea*0.5* LeakageFactor;
		
		elseif AHUAirflowMethod= IndoorUnitMethod then
		AHUAirflow=IndoorAirUnitCoolingCapacityton*350*LeakageFactor >>

11	Actual Duct Leakage Rate from Leakage Test Measurement (cfm)	<<user input: numeric xxx.x>>
12	Air-Handling Unit Manufacturer Name	<<user input, text, maximum 200 characters>>
13	Air-Handling Unit Model Number	<<user input, text, maximum 200 characters>>
14	Compliance Statement:	<<if B11 measured leakage rate is ≤ to B10 target allowable leakage rate: "system passes leakage test"; else if measured leakage rate is > target allowable leakage rate: "system fails leakage test">>
15	Notes: <<if LMCI-MCH-20 row B02, B03, B04 and B05 ≠ LMCV-MCH-20 row B01=2, B03, B04 and B05, then display “The installed cooling capacity, heating capacity or CFA served does not match the Installation Certificate”, elseif LMCI-MCH-20 row B02, B03, B04 and B05 = LMCV-MCH-20 row B02, B03, B024 and B05, then display “ “; if LMCI-MCH-20 row B12 and B13 ≠ LMCV-MCH-20 row B12 and B13, then display “The Air-Handling Unit Manufacturer Name or Model Number does not match the Installation Certificate”, elseif LMCI-MCH-20 row B12 and B13 = LMCV-MCH-20 row B12 and B13, then display “ “>>	

For information and data collection only. Not valid until registered with an ECC provider

B4. Duct Leakage Diagnostic Test for Complete Replacement or Altered Duct System

<< If A07= Replacement or Alteration this section is required Else only display Section title and this note:
This section does not apply to this project.>>

01	Air-Handler Unit Airflow (AHU Airflow) Determination Method	<<pick one from list: DefaultAirflowMethod; CoolingSystemMethod; HeatingSystemMethod; MeasuredAirflowMethod; IndoorUnitMethod>>
02	Condenser Nominal Cooling Capacity (ton)	<<if B01 = CoolingSystemMethod, then user input is numeric x.xx; else =N/A>>
03	Indoor Unit Nominal Cooling Capacity (ton)	<< if B01 = IndoorUnitMethod, then user input is either numeric x.xx, else =N/A>>
04	Heating Capacity (kBtu/h)	<< if B01 = HeatingSystemMethod, then user input is numeric xxx.x; else =N/A >>
05	Conditioned Floor Area Served by this HVAC System (ft^2)	<<if B01 = DefaultAirflowMethod, user input is numeric xx,xxx; else = N/A (should be consistent with LMCI-MCH-20)>>
06	Measured AHU Airflow (cfm)	<<if B01 = MeasuredAirflowMethod, then user enter numeric x,xxx, else =N/A>>
07	Duct Leakage Test Conditions	<<Auto filled field: TestFinal (this is the only allowable test condition for Replacement/Alteration)>>
08	Duct Leakage Test Method	<<user pick one from list: LeakageToOutside; TotalLeakage>>

09	Leakage Factor	<<calculated field:
		if TotalLeakage and MultiFamily and TestFinal and Replacement then LeakageFactor=0.12;
		
		elseif LeakageToOutside and MultiFamily and TestFinal and Replacement then LeakageFactor=0.06;
		
		elseif TotalLeakage and TestFinal and Alteration then LeakageFactor=0.15 ;
		
		elseif LeakageToOutside and TestFinal and Alteration then LeakageFactor=0.10 ;
		
		else error message if invalid entries for arguments>>

10	Calculated Target Allowable Duct Leakage Rate (cfm)	<<calculated field: numeric xxx:
		if AHUAirflowMethod= DefaultAirflowMethod then AHUAirflow=ZonedCondFloorArea*0.5* LeakageFactor;
		
		elseif AHUAirflowMethod= CoolingSystemMethod and A09 = no, then AHUAirflow=CondenserNomCoolCapacityTon*400* LeakageFactor;
		
		elseif AHUAirflowMethod = CoolingSystemMethod and A09=yes, then value=CondenserNomCoolCapacityTon *250*LeakageFactor;
		
		elseif AHUAirflowMethod= HeatingSystemMethod then AHUAirflow=HeatingCapacityKbtuh*21.7* LeakageFactor;
		
		elseif AHUAirflowMethod= MeasuredAirflowMethod then AHUAirflow= Measured AHUAirflow * LeakageFactor;
		
		elseif AHUAirflowMethod= IndoorUnitMethod then AHUAirflow=IndoorAirUnitCoolingCapacityton*350*LeakageFactor>>

11	Actual Duct Leakage Rate from Leakage Test Measurement (cfm)	<<user input: numeric xxx.x>>
12	Compliance Statement:	<<if B11 measured leakage rate is ≤ to B10 target allowable leakage rate: "system passes leakage test"; else if measured leakage rate is > target allowable leakage rate: "system fails leakage test">>

For information and data collection only. Not valid until registered with an ECC provider

13	Notes: <<if LMCI-MCH-20 row B02, B03, B04 and B05 ≠ LMCV-MCH-20 row B02, B03, B04 and B05, then display “The installed cooling capacity, heating capacity or CFA served does not match the Installation Certificate”, elseif LMCI-MCH-20 row B02, B03 and B04, B05 = LMCV-MCH-20 row B02, B03, B04 and B05, then display “ “>>

B5. Duct Leakage Diagnostic Test for Smoke Test or Alteration Using Smoke Test

<< If A07= Replacement using Smoke Test or Alteration using Smoke Test this section is required Else only display Section title and this note: *This section does not apply to this project.*>>

01	Air-Handling Unit Airflow (AHU Airflow) Determination Method	<<pick one from list: DefaultAirflowMethod; CoolingSystemMethod; HeatingSystemMethod; MeasuredAirflowMethod; IndoorUnitMethod>>
02	Condenser Nominal Cooling Capacity (ton)	<< if B01 = CoolingSystemMethod, then user input is numeric x.xx; else =N/A>>
03	Indoor Unit Nominal Cooling Capacity (ton)	<< if B01 = IndoorUnitMethod, then user input is either numeric x.xx, else =N/A>>
04	Heating Capacity (kBtu/h)	<< if B01 = HeatingSystemMethod, then user input is numeric xxx.x; else =N/A >>
05	Conditioned Floor Area Served by this HVAC System (ft^2)	<< if B01 = DefaultAirflowMethod, user input is numeric xx,xxx; else = N/A>>
06	Measured AHU Airflow (cfm)	<< if B01 = MeasuredAirflowMethod, then user enter numeric x,xxx, else =N/A>>
07	Duct Leakage Test Conditions	<<Auto filled field: TestFinal (this is the only allowable test condition for Replacement/Alteration using Smoke Test)>>
08	Duct Leakage Test Method	<< Auto filled field: TotalLeakage (this is the only allowable test method for Replacement/Alteration using Smoke Test)>>

09	Leakage Factor	<<calculated field:
		if TotalLeakage and MultiFamily and TestFinal and Replacement using Smoke Test then LeakageFactor=0.12;
		
		elseif TotalLeakage and TestFinal and Alteration using Smoke Test then LeakageFactor=0.15;
		
		else provide applicable error message if invalid entries for arguments>>

10	Calculated Target Allowable Duct Leakage Rate (cfm)	<<calculated field: numeric xxx:
		if AHUAirflowMethod=DefaultAirflowMethod then AHUAirflow=ZonedCondFloorArea*0.5* LeakageFactor;
		
		elseif AHUAirflowMethod= CoolingSystemMethod and A09=no, then AHUAirflow=CondenserNomCoolCapacityTon*400* LeakageFactor;
		
		elseif AHUAirflowMethod = CoolingSystemMethod and A09=yes, then value=CondenserNomCoolCapacityTon *250*LeakageFactor;
		
		elseif AHUAirflowMethod= HeatingSystemMethod then AHUAirflow=HeatingCapacityKbtuh*21.7* LeakageFactor;
		
		elseif AHUAirflowMethod= MeasuredAirflowMethod then AHUAirflow= Measured AHUAirflow * LeakageFactor;
		
		elseif AHUAirflowMethod= IndoorUnitMethod then AHUAirflow=IndoorAirUnitCoolingCapacityton*350*LeakageFactor >>

11	Actual Duct Leakage Rate from Leakage Test Measurement (cfm)	<<user input: numeric xxx.x>>
12	Compliance Statement:	<<if B11 measured leakage is ≤ to B10 target allowed, then display message: "system passes - system complies with Allowable Duct Leakage Rate criterion"; else if B11 measured leakage rate is greater than B10 target allowable leakage rate then user selects one of following two results: **”System Fails the smoke test of an altered HVAC system in an existing building” or **”System passes using smoke test of an altered HVAC system in an existing building • No visible smoke exits the accessible portions of the duct system.

For information and data collection only. Not valid until registered with an ECC provider

07		• Smoke is only emanating from air-handling unit (AHU) cabinet and non-accessible portions of the duct system.” Note – Accessible is defined as having access thereto, but which first may require removal or opening of access panels, doors, or moving similar obstructions. If access to the ducts requires an object to be demolished or deconstructed then sealing of those ducts is not required.">>
13	Notes: <<if LMCI-MCH-20 row B02, B03, B04 and B05 ≠ LMCV-MCH-20 row B02, B03, B04 and B05, then display “The installed cooling capacity, heating capacity or CFA served does not match the Installation Certificate”, elseif LMCI-MCH-20 row B02, B03, B04 and B05 = LMCV-MCH-20 row B02, B03, B04 and B05, then display “ “>>	

C. Ducts Located in Garage Spaces

<<if A08 = yes, then show table, else display the Section Does Not Apply message>>

01	Duct Leakage Test Method	<<default value = TotalLeakage (this is the only method allowed>>
02	Leakage Factor	<<default value = 0.06>>
03	Air-Handling Unit Airflow (AHU Airflow) Determination Method	<<auto filled from B01>>
04	Measured AHU Airflow (cfm)	<< auto filled from B06>>

05	Calculated Target Allowable Duct Leakage Rate (cfm)	<<calculated field: numeric xxx:
		if AHUAirflowMethod= CoolingSystemMethod then AHUAirflow=CondenserNomCoolCapacityTon*400* 0.06;
		
		elseif AHUAirflowMethod= HeatingSystemMethod then AHUAirflow=HeatingCapacityKbtuh*21.7* 0.06;
		
		elseif AHUAirflowMethod= MeasuredAirflowMethod then AHUAirflow= Measured AHUAirflow * 0.06
		
		elseif AHUAirflowMethod= IndoorUnitMethod then AHUAirflow=IndoorAirUnitCoolingCapacityton*350*0.06>>

06	Actual Duct Leakage Rate from Leakage Test Measurement (cfm)	<< auto filled from B11>>
07	Compliance Statement:	<<if measured leakage rate is ≤ to target allowable leakage rate, then display message: "Ducts in garage passes – overall system leakage complies"; else if measured leakage rate is > target allowable leakage rate then display message: "Ducts in garage passes using smoke test of an altered HVAC system in an existing building • No visible smoke exits the accessible portions of the ducts in the garage”>>

D. Additional Requirements For Compliance

01	System was tested in its normal operation condition. No temporary taping allowed.
02	Outside air (OA) duct connections to the central forced air duct system shall not be sealed/taped off during duct leakage testing. OA ducts used for Central Fan Integrated (CFI) Indoor Air Quality ventilation systems, or Central Fan Ventilation Cooling Systems, that utilize dampers that open only when OA is required and automatically close when OA is not required, may configure the OA damper to the closed position during duct leakage testing.
03	All supply and return register boots were sealed to the drywall.
04	Building cavities were not used as plenums, or platform returns, in lieu of ducts.
05	If cloth backed tape was used it was covered with Mastic and draw bands.
06	All connection points between the air handler and the supply and return plenums are completely sealed.
07	For completely new systems visual inspection at final construction stage (applicable if system was tested at rough-in): For all supply and return registers, verify that the spaces between the register boot and the interior finishing wall are properly sealed.

For information and data collection only. Not valid until registered with an ECC provider

08	For completely new systems visual inspection at final construction stage (applicable if system was tested at rough-in): If the house rough-in duct leakage test was conducted without an air handler installed, inspect the connection points between the air handler and the supply and return plenums to verify that the connection points are properly sealed.	
09	For completely new systems visual inspection at final construction stage (applicable if system was tested at rough-in): Inspect all joints to ensure that no cloth backed rubber adhesive duct tape is used.	
10	For Duct Systems with Low Leakage Air-Handling Unit (LLAHU): The Low Leakage Air-handling Unit Model identified on this compliance document is included in the list of certified Low Leakage Air-Handling Units published on the Energy Commission Website at: https://www.energy.ca.gov/rules-and-regulations/building-energy-efficiency/manufacturer-certification-building-equipment/low	
11	For Replacement or Alteration Duct Systems: If the system complies using the Smoke Test method, the smoke test was conducted in accordance with the requirements of Reference Residential Appendix RA3.1.4.3.6. Systems that comply using the smoke test shall not be included in sample groups for ECC verification compliance.	
12	Verification Status:	<<user pick from list: *** Pass - all applicable requirements are met; or *** Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or *** All n/a - This entire table is not applicable
13	Correction Notes:	<<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.		

E. Determination of ECC Verification Compliance All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance.	
01	<<if Compliance Statement in Section B1 or B2 or B3 or B4 or B5 =System Passes, and D12≠Fail, then display: "Complies: All specified verification protocol requirements on this document are met"; else display: "Does not comply: One or more specified verification protocol requirements on this document are not met" >>

For information and data collection only. Not valid until registered with an ECC provider